Year 2 Pure Mathematics

Chapter 11

Integration

Practice Booklet

Contents

11.1 Integrating standard functions ..... book p. 90
11.2 Integrating f($ax + b$) ..... book p. 91
11.3 Using trigonometric identities ..... book p. 92
11.4 Reverse chain rule ..... book p. 92
11.5 Integration by substitution ..... book p. 93
11.6 Integration by parts ..... book p. 94
11.7 Partial fractions ..... book p. 95
11.8 Finding areas ..... book p. 96
11.9 The trapezium rule ..... book p. 98
11.10 Solving differential equations ..... book p. 100
11.11 Modelling with differential equations ..... book p. 101
Problem solving: Set A ..... book p. 103
Problem solving: Set B ..... book p. 104
Selected exam question bank questions .....
Answers .....

11.1 Integrating standard functions

Practice Book, printed page 90.

1. Integrate the following with respect to x:

a. $e^x + 2\sin x$

b. $4\cos x - \dfrac{3}{x^2}$

c. $\dfrac{5}{x} - 2e^x$

d. $\sqrt{x} + \dfrac{3}{4}\sin x$

e. $11x^2 + \dfrac{1}{2}e^x - \dfrac{6}{x}$

f. $3\cos x - \dfrac{1}{\sqrt{x}}$

Hint

Use the following standard integrals to help:

$$\int x^n \, \mathrm{d}x = \frac{x^{n+1}}{n+1} + c$$

$$\int e^x \, \mathrm{d}x = e^x + c$$

$$\int \frac{1}{x} \, \mathrm{d}x = \ln|x| + c$$

$$\int \sin x \, \mathrm{d}x = -\cos x + c$$

$$\int \cos x \, \mathrm{d}x = \sin x + c$$

2. Find the following integrals.

a. $\displaystyle\int \frac{1}{\sin^2 x}\,\mathrm{d}x$

b. $\displaystyle\int \frac{\sin x}{\cos^2 x}\,\mathrm{d}x$

c. $\displaystyle\int \frac{\cos x}{\sin^2 x}\,\mathrm{d}x$

d. $\displaystyle\int \frac{4}{\cos^2 x}\,\mathrm{d}x$

e. $\displaystyle\int \frac{4}{1-\cos^2 x}\,\mathrm{d}x$

f. $\displaystyle\int \frac{8\cos x}{\cos^2 x - 1}\,\mathrm{d}x$

Hint

Use the following standard integrals to help:

$$\int \sec^2 x\,\mathrm{d}x = \tan x + c$$

$$\int \operatorname{cosec} x \cot x\,\mathrm{d}x = -\operatorname{cosec} x + c$$

$$\int \operatorname{cosec}^2 x\,\mathrm{d}x = -\cot x + c$$

$$\int \sec x \tan x\,\mathrm{d}x = \sec x + c$$

3. Find the following integrals.

a. $\displaystyle\int \sec x(\sec x + \tan x)\,\mathrm{d}x$

b. $\displaystyle\int \frac{7x^2 - 5}{x}\,\mathrm{d}x$

c. $\displaystyle\int \operatorname{cosec}^2 x(3e^x \sin^2 x - 5)\,\mathrm{d}x$

d. $\displaystyle\int \frac{x^2 - x + 1}{x}\,\mathrm{d}x$

Hint

Simplify each expression so it forms one or more standard integral.

4. Evaluate the following. Give your answers as exact values.

a. $\displaystyle\int_{\ln 5}^{\ln 20} e^x\,\mathrm{d}x$

b. $\displaystyle\int_{2}^{5} \frac{1 - 3x^3}{x}\,\mathrm{d}x$

c. $\displaystyle\int_{0}^{\frac{\pi}{3}} \left(\frac{\sin x}{\cos^2 x} + 4e^x\right)\mathrm{d}x$

d. $\displaystyle\int_{0}^{\frac{\pi}{4}} \sec^2 x(4 - \cos x \tan x)\,\mathrm{d}x$

e. $\displaystyle\int_{\ln 2}^{\ln 6} (e^x + e^{2x})\,\mathrm{d}x$

5. E Find the exact value of $\displaystyle\int_4^9 \frac{x(1+x)}{x^2}\,\mathrm{d}x$, leaving your answer in the form $a + \ln b$, where a and b are rational numbers. (4 marks)

6. E/P $\displaystyle\int_3^9 \frac{p}{x}\,\mathrm{d}x = \ln 81$, where p is a positive constant.
Find the value of p. (4 marks)

7. E Find the exact value of $\displaystyle\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sec^2 x(\cot^2 x - 3e^x \cos^2 x - 1)\,\mathrm{d}x$ (5 marks)

8. E/P Given that a is a real constant, and that

$$\int_{2a}^{5a} \frac{2x+3}{x}\,\mathrm{d}x = \ln 1000$$

show that $a = \ln k$, where k is a constant to be found. (5 marks)

11.2 Integrating f$(ax + b)$

Practice Book, printed page 91.

1.

a. Differentiate:

i. $(4x-5)^4$

ii. $(3-2x)^6$

b. Find:

i. $\displaystyle\int 16(4x-5)^3\,\mathrm{d}x$

ii. $\displaystyle\int (4x-5)^3\,\mathrm{d}x$

iii. $\displaystyle\int (-12(3-2x)^5)\,\mathrm{d}x$

iv. $\displaystyle\int 2(3-2x)^5\,\mathrm{d}x$

Hint

Integration is the reverse process of differentiation, so use your answers to part a to help.

2. Integrate the following with respect to x.

a. $6e^{3x}$

b. $\cos(3x+2)$

c. $(4x+1)^5$

d. $\operatorname{cosec} 4x \cot 4x$

e. $\sec^2(8x-1)$

f. $\operatorname{cosec}^2(1-5x)$

g. $\dfrac{5}{2x-3}$

h. $11e^{4-x}$

Hint

In general,

$$\int \mathrm{f}'(ax+b)\,\mathrm{d}x = \frac{1}{a}\mathrm{f}(ax+b) + c$$

3. Find the following integrals.

a. $\displaystyle\int \left(\frac{1}{(4x-3)^2} - (4x-3)^2\right)\mathrm{d}x$

b. $\displaystyle\int \sec^2 3x \left(\frac{1}{2}e^{5x}\cos^2 3x - 2\right)\mathrm{d}x$

c. $\displaystyle\int \left(e^{2x} - \frac{1}{e^{2x}}\right)^2 \mathrm{d}x$

d. $\displaystyle\int \frac{5 - 3\cos 2x}{\sin^2 2x}\,\mathrm{d}x$

Hint

First simplify each expression and then apply the integration rule in question 2 to each term separately.

4. Find the exact value of:

a. $\displaystyle\int_{-2}^{0} \frac{4}{6 - 4x}\,\mathrm{d}x$

b. $\displaystyle\int_{-\frac{\pi}{8}}^{\frac{\pi}{8}} \operatorname{cosec}^2 2x(1 - \cos 2x)\,\mathrm{d}x$

c. $\displaystyle\int_{0}^{\ln 4} (e^x - 4)^2\,\mathrm{d}x$

d. $\displaystyle\int_{\frac{5\pi}{8}}^{\frac{3\pi}{4}} \frac{1}{\sin^2(2\pi - x)}\,\mathrm{d}x$

Hint

Integrate each expression and then apply the limits to evaluate the integral.

5. E/P Given that a is a positive constant and $\displaystyle\int_{1}^{a} (2x + 1)^2\,\mathrm{d}x = 117$, find the value of a. (4 marks)

6. E/P $\displaystyle\int_{1}^{4} \frac{k}{2x - 1}\,\mathrm{d}x = \ln 25$, where k is a real constant.
Find the value of k. (4 marks)

7. E/P Find the exact value of $\displaystyle\int_{\ln 2}^{\ln 8} (e^x + 2)^2\,\mathrm{d}x$, giving your answer in the form $p + \ln q$, where p and q are integers. (4 marks)

8. E/P Given that $\displaystyle\int_{\frac{\pi}{6k}}^{\frac{\pi}{3k}} \frac{\pi}{2}\cos kx\,\mathrm{d}x = \frac{9\pi}{4}(1 - \sqrt{3})$, where k is a real constant, find the value of k. (5 marks)

11.3 Using trigonometric identities

Practice Book, printed page 92.

1. Integrate the following with respect to x.

a. $\sin^2 x$

b. $3\cot^2 x$

c. $(1 + \cos x)^2$

d. $\dfrac{\sin^2 x}{\cos^2 x}$

Hint

Use a trigonometric identity to transform each expression into a sum of standard integrals. ⟵ Sections 6.4, 7.3

2. Find the exact value of:

a. $\displaystyle\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{\cos 2x}{\sin^2 x}\,\mathrm{d}x$

b. $\displaystyle\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} (\sin x - \operatorname{cosec} x)^2 \, \mathrm{d}x$

c. $\displaystyle\int_{\frac{\pi}{3}}^{\frac{2\pi}{3}} \frac{(1-\cos x)^2}{\sin^2 x} \, \mathrm{d}x$

d. $\displaystyle\int_{0}^{\frac{\pi}{4}} (\sec x - \tan x)^2 \, \mathrm{d}x$

Hint

In part a, use $\cos 2A \equiv 1 - 2\sin^2 A$ together with the standard result for integrating $\operatorname{cosec}^2 x$. ⟵ Section 7.3

3.

a. Using the addition formula, expand:

i. $\cos(5x + 2x)$

ii. $\cos(5x - 2x)$

b. By adding parts a i and a ii together, show that $\cos 7x + \cos 3x = 2\cos 5x \cos 2x$

c. Use your answer to part b to find $\displaystyle\int 4\cos 5x \cos 2x \, \mathrm{d}x$.

Hint

$\cos(A + B) \equiv \cos A \cos B - \sin A \sin B$
$\cos(A - B) \equiv \cos A \cos B + \sin A \sin B$
⟵ Section 7.1

4. E/P Show that $\displaystyle\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \cos^2 x \, \mathrm{d}x = \frac{\pi}{6} - \frac{\sqrt{3}}{8}$ (4 marks)

5. E/P Show that $\displaystyle\int_{\frac{\pi}{12}}^{\frac{\pi}{6}} \tan^2 2x \, \mathrm{d}x = \frac{\sqrt{3}}{3} - \frac{\pi}{12}$ (5 marks)

6. E/P

a. By considering $\sin(6x+x)$ and $\sin(6x-x)$, or otherwise, show that $\sin 7x + \sin 5x = 2\sin 6x \cos x$ (3 marks)

b. Hence find $\displaystyle\int 3\sin 6x \cos x \, \mathrm{d}x$. (3 marks)

11.4 Reverse chain rule

Practice Book, printed page 92.

1.

a. Differentiate:

i. $(x^2 - 4)^3$

ii. $(x^2 + 4x - 5)^4$

b. Find:

i. $\displaystyle\int 6x(x^2 - 4)^2 \, \mathrm{d}x$

ii. $\displaystyle\int 15x(x^2 - 4)^2 \, \mathrm{d}x$

iii. $\displaystyle\int -8(x + 2)(x^2 + 4x - 5)^2 \, \mathrm{d}x$

Hint

Integration is the reverse process of differentiation, so use your answers to part a to help.

2. Find the following integrals.

a. $\int 6x(x^2-7)^{\frac{1}{2}}\,\mathrm{d}x$

b. $\int 5\sin 2x\, e^{\cos 2x}\,\mathrm{d}x$

c. $\int (6x^2+8)(4x^3+16x-5)^{\frac{3}{2}}\,\mathrm{d}x$

d. $\int \sec^2 x\sqrt{1+\tan x}\,\mathrm{d}x$

e. $\int \operatorname{cosec}^2 3x\, e^{\cot 3x}\,\mathrm{d}x$

f. $\int \cos^4 5x \sin 5x\,\mathrm{d}x$

Hint

To integrate an expression in the form $\int k\mathrm{f}'(x)(\mathrm{f}(x))^n\,\mathrm{d}x$, try differentiating $(\mathrm{f}(x))^{n+1}$, then adjust for a constant if necessary.

3. Find the following integrals.

a. $\int \dfrac{4x}{2-x^2}\,\mathrm{d}x$

b. $\int \dfrac{\sin x}{4-3\cos x}\,\mathrm{d}x$

c. $\int \dfrac{6x^2-9}{14+9x-2x^3}\,\mathrm{d}x$

d. $\int \dfrac{\sin 2x}{\cos^2 x}\,\mathrm{d}x$

e. $\int \dfrac{\frac{1}{2}\sin x\cos x}{4-\cos 2x}\,\mathrm{d}x$

f. $\int \dfrac{e^{-3x}}{1+e^{-3x}}\,\mathrm{d}x$

Hint

To integrate an expression in the form $\int k\dfrac{\mathrm{f}'(x)}{\mathrm{f}(x)}\,\mathrm{d}x$, try differentiating $\ln(\mathrm{f}(x))$, then adjust for a constant if necessary.

4. Find the exact value of:

a. $\int_{\frac{3\pi}{8}}^{\frac{\pi}{2}} \sin^2 4x\cos 4x\,\mathrm{d}x$

b. $\int_0^{\frac{\pi}{9}} \dfrac{\sec^2 3x\tan 3x}{\sec^2 3x}\,\mathrm{d}x$

c. $\int_{\ln 2}^{\ln 3} \dfrac{6e^{3x}}{e^{3x}+1}\,\mathrm{d}x$

d. $\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \operatorname{cosec} 3x\cot 3x\, e^{\operatorname{cosec} 3x}\,\mathrm{d}x$

Hint

Decide carefully whether each expression is in the form $kf'(x)(f(x))^n$ or $k\dfrac{f'(x)}{f(x)}$.
Then integrate and apply the limits to evaluate each integrand.

5. E/P Evaluate $\displaystyle\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} (1+\cot x)^3 \operatorname{cosec}^2 x \, dx$, leaving your answer in the form $a+b\sqrt{3}$. (5 marks)

6. E/P

a. Show that $\sin^5 x \equiv \sin x - 2\sin x \cos^2 x + \sin x \cos^4 x$ (3 marks)

b. Hence find $\displaystyle\int \sin^5 x \, dx$. (3 marks)

7. E Evaluate $\displaystyle\int_0^4 (x+1)\sqrt{2x^2+4x}\, dx$, leaving your answer in the form $k\sqrt{3}$. (3 marks)

8. E Find the exact value of $\displaystyle\int_0^{\frac{\pi}{8}} \sin 4x (e^{\cos 4x})\, dx$. (4 marks)

9. E Find the exact value of $\displaystyle\int_0^{\frac{\pi}{4}} \frac{\sec^2 x}{(3-\tan x)^4}\, dx$, leaving your answer as a simplified fraction. (5 marks)

10. E/P Given that $\displaystyle\int_1^k \frac{x}{x^2+3}\, dx = \ln\frac{3}{2}$ and $k>0$, find the exact value of k. (4 marks)

11.5 Integration by substitution

Practice Book, printed page 93.

1. Complete the following steps to find $\displaystyle\int 2x(x+2)^4\, dx$ using integration by substitution.

a. Given $u=x+2$, find $\dfrac{du}{dx}$

b. Given $u=x+2$, write $2x$ in terms of u.

c. Use your answers to parts a and b to show that $\displaystyle\int 2x(x+2)^4\, dx = \int 2(u-2)u^4\, du$.

d. Show that $\displaystyle\int 2(u-2)u^4\, du = \frac{1}{3}u^6 - \frac{4}{5}u^5 + c$

e. Use the substitution in part a to write your answer in part d in terms of x.

2. Use the given substitution to find:

a. $\displaystyle\int x(x-8)^4\, dx$; $u=x-8$

b. $\displaystyle\int \sin x \cos x (1+\cos x)^5\, dx$; $u=1+\cos x$

c. $\displaystyle\int \operatorname{cosec}^2 x \cot x \sqrt{2+\cot x}\, dx$; $u=2+\cot x$

d. $\displaystyle\int \frac{\sqrt{x^2+1}}{x}\, dx$; $u=\sqrt{x^2+1}$

Hint

Use $\dfrac{du}{dx}$ to replace dx with an expression involving du. Then write the integrand in terms of u and integrate. Remember to use the substitution to write your final answer in terms of x.

3. Use the given substitution to evaluate:

a. $\int_{\frac{1}{2}}^{\frac{5}{4}} x\sqrt{4x-1}\,\mathrm{d}x;\ u = 4x-1$

b. $\int_{16}^{25} \frac{4}{\sqrt{x}(\sqrt{x}-9)}\,\mathrm{d}x;\ u = \sqrt{x}$

c. $\int_{\frac{2\pi}{3}}^{\frac{3\pi}{4}} \frac{\sin x}{\cos^3 x}\,\mathrm{d}x;\ u = \cos x$

d. $\int_{0}^{\frac{\pi}{2}} \frac{\sin 2x}{1+\sin x}\,\mathrm{d}x;\ u = \sin x$

4. E/P Using the substitution $u = 1+x^2$ or otherwise, find the exact value of $\int_0^2 x^3\sqrt{1+x^2}\,\mathrm{d}x$.(6 marks)

5. E/P Using the substitution $u = e^x$, or otherwise, find the exact value of $\int_{\ln 2}^{\ln 6} \frac{1}{1+e^x}\,\mathrm{d}x$, leaving your answer in the form $\ln a$, where a is a rational number to be found. (6 marks)

6. E

a. Using the substitution $x = \tan\theta$, or otherwise, show that $\int \frac{1}{1+x^2}\,\mathrm{d}x = \arctan x + c$.(4 marks)

b. Hence evaluate $\int_{\frac{\sqrt{3}}{3}}^{1} \frac{1}{1+x^2}\,\mathrm{d}x$. (2 marks)

7. E/P Using the substitution $u^2 = 2x+1$ or otherwise, show that the exact value of $\int_0^4 \frac{5x}{\sqrt{2x+1}}\,\mathrm{d}x$ is $\frac{370\sqrt{3}-20}{21}$ (6 marks)

8. E/P By using a suitable substitution, or otherwise, find $\int \frac{2\sin 2x}{1+\sin x}\,\mathrm{d}x$. (5 marks)

11.6 Integration by parts

Practice Book, printed page 94.

1. Find the following integrals.

a. $\int x\cos x\,\mathrm{d}x$

b. $\int x\sec^2 x\,\mathrm{d}x$

c. $\int xe^{2x}\,\mathrm{d}x$

d. $\int x\,\mathrm{cosec}^2\,4x\,\mathrm{d}x$

e. $\int xe^{-4x}\,\mathrm{d}x$

f. $\int x\sin 5x\,\mathrm{d}x$

Hint

$\int u\frac{\mathrm{d}v}{\mathrm{d}x}\,\mathrm{d}x = uv - \int v\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x$

In part a, let $u = x$ and $\frac{\mathrm{d}v}{\mathrm{d}x} = \cos x$.

2. Find the following integrals.

a. $\int x \ln x \, \mathrm{d}x$

b. $\int x^3 \ln x \, \mathrm{d}x$

c. $\int \ln x \, \mathrm{d}x$

Hint

In part a, let $u = \ln x$ and $\dfrac{\mathrm{d}v}{\mathrm{d}x} = x$.

In part c, write $\int \ln x \, \mathrm{d}x$ as $\int 1 \times \ln x \, \mathrm{d}x$.

3. Find the following integrals.

a. $\int x^2 \sin 2x \, \mathrm{d}x$

b. $\int x^2 e^{-3x} \, \mathrm{d}x$

c. $\int 2x^2(1+x)^3 \, \mathrm{d}x$

Hint

You will need to apply integration by parts twice to answer each part of this question.

4. Evaluate:

a. $\displaystyle\int_0^{\frac{\pi}{4}} x \sin x \, \mathrm{d}x$

b. $\displaystyle\int_2^5 x\sqrt{6-x} \, \mathrm{d}x$

c. $\displaystyle\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} x \operatorname{cosec}^2 x \, \mathrm{d}x$

d. $\displaystyle\int_0^{\frac{\pi}{6}} x \sec^2 2x \, \mathrm{d}x$

e. $\displaystyle\int_0^{\frac{\pi}{4}} x^2 \cos 4x \, \mathrm{d}x$

f. $\displaystyle\int_0^3 x^3 e^{-x} \, \mathrm{d}x$

5. E Find the exact value of $\displaystyle\int_{\frac{1}{2}}^2 6x \ln 4x \, \mathrm{d}x$. (5 marks)

6. E Show that $\displaystyle\int_1^2 x^2 e^{2x} \, \mathrm{d}x = \frac{e^2}{4}(5e^2 - 1)$. (6 marks)

7. E

a. Use integration to find $\displaystyle\int x \sin \pi x \, \mathrm{d}x$. (3 marks)

b. Hence, or otherwise, evaluate $\displaystyle\int_{\frac{3}{2}}^{\frac{5}{2}} x^2 \cos \pi x \, \mathrm{d}x$, leaving your answer as an exact value in terms of π. (5 marks)

8. E Find $\displaystyle\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} 2x^2 \operatorname{cosec}^2 x \cot x \, \mathrm{d}x$, leaving your answer in exact form. (5 marks)

9. E/P

a. Find $\int x \cos 2x \, \mathrm{d}x$. (3 marks)

b. Hence, using a suitable trigonometric identity, find $\int x \sin^2 x \, \mathrm{d}x$. (3 marks)

11.7 Partial fractions

Practice Book, printed page 95.

1. $\dfrac{8x-4}{(x+2)(x-4)} \equiv \dfrac{A}{x+2} + \dfrac{B}{x-4}$

a. Find the values of the constants A and B.

b. Hence find $\displaystyle\int \frac{8x-4}{(x+2)(x-4)} \, \mathrm{d}x$.

Hint

Write the integral using the expression you found in part a, then use integrate each term using

$$\int \frac{1}{ax+b} \, \mathrm{d}x = \frac{1}{a} \ln |ax+b| + c$$

2. Use partial fractions to integrate:

a. $\displaystyle\int \frac{4x+17}{(x+5)(x+2)} \, \mathrm{d}x$

b. $\displaystyle\int \frac{7x+6}{(x+3)(2-x)} \, \mathrm{d}x$

c. $\displaystyle\int \frac{x+17}{(x-1)(x+2)} \, \mathrm{d}x$

d. $\displaystyle\int \frac{6}{(x-4)(2x-1)} \, \mathrm{d}x$

Hint

First rewrite each expression using partial fractions. ⟵ Section 1.3

3. $\mathrm{f}(x) = \dfrac{-2x^2+14}{(x+4)(2x-1)}$, $x \in \mathbb{R}$, $x \neq -4$, $x \neq \dfrac{1}{2}$

a. Use algebraic long division to show that $\mathrm{f}(x) \equiv \dfrac{7x+10}{(x+4)(2x-1)} - 1$.

b. Use partial fractions to find values A and B such that $\mathrm{f}(x) \equiv \dfrac{A}{x+4} + \dfrac{B}{2x-1} - 1$.

c. Hence find $\displaystyle\int \frac{-2x^2+14}{(x+4)(2x-1)} \, \mathrm{d}x$.

4.

a. Show that $\dfrac{16x^2-37x+17}{(4-x)(2x-3)^2} = \dfrac{5}{4-x} + \dfrac{2}{2x-3} - \dfrac{1}{(2x-3)^2}$

b. Hence find $\displaystyle\int \frac{16x^2-37x+17}{(4-x)(2x-3)^2} \, \mathrm{d}x$.

Hint

$2x-3$ is a repeated factor, so the partial fraction expansion includes separate denominators of $(2x-3)$ and $(2x-3)^2$.
⟵ Section 1.4

5. E $\mathrm{f}(x) = \dfrac{3x+15}{(x-1)(x+2)}$, $x > 1$

a. Express $\mathrm{f}(x)$ in partial fractions. (3 marks)

b. Find $\displaystyle\int_2^3 \mathrm{f}(x)\,\mathrm{d}x$, giving your answer in the form $\ln k$, where k is a rational constant. (5 marks)

6. E $\mathrm{f}(x) = \dfrac{2x^2-5x+5}{2x^2-5x+2}$, $x > 1$

a. Given that $\mathrm{f}(x) = \dfrac{A}{x-2} + \dfrac{B}{2x-1} + C$, find the values of the constants A, B and C. (4 marks)

b. Hence find the exact value of $\displaystyle\int_3^5 \frac{2x^2-5x+5}{2x^2-5x+2}\,\mathrm{d}x$, writing your answer in the form $a + \ln b$, where a and b are rational constants to be found. (5 marks)

7. E $\mathrm{f}(x) = \dfrac{12x+6}{(2x+1)^2(x+3)}$, $x > 0$

a. Express $\mathrm{f}(x)$ in partial fractions. (5 marks)

b. Find $\displaystyle\int_1^2 \mathrm{f}(x)\,\mathrm{d}x$, giving your answer in the form $a\ln b$, where a and b are rational constants. (5 marks)

8. E $\mathrm{f}(x) = \dfrac{x^2+4}{x^2-1}$, $x > 1$

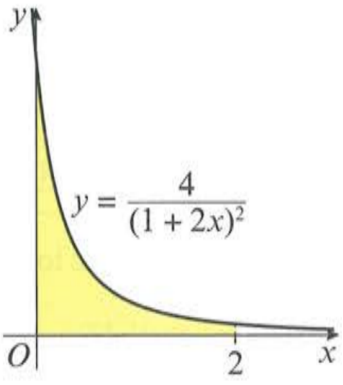
a. Given that $\mathrm{f}(x) = A + \dfrac{B}{x+1} + \dfrac{C}{x-1}$, find the values of the constants A, B and C. (5 marks)

b. Find $\displaystyle\int_2^4 \mathrm{f}(x)\,\mathrm{d}x$, giving your answer in the form $a + b\ln c$, where a, b and c are rational constants. (5 marks)

11.8 Finding areas

Practice Book, printed page 96.

1. The diagram shows part of the curve $y = \dfrac{4}{(1+2x)^2}$



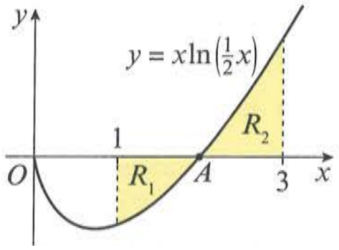
The shaded region is bounded by the curve, the coordinate axes and the line with equation $x = 2$. Find the exact area of the shaded region.

Hint

The area under the curve between $x = 0$ and $x = 2$ is given by $\displaystyle\int_0^2 \frac{4}{(1+2x)^2}\,\mathrm{d}x$

2. The diagram shows a sketch of part of the curve with equation

$$y = x \ln\left(\frac{1}{2}x\right), \quad x \geq 0$$



a. Verify that A has coordinates $(2, 0)$.

b. Show that the exact area of the shaded region R_1 is $\frac{3}{4} - \frac{1}{2}\ln 2$

c. Show that the total shaded area is $\frac{9}{2}\ln\frac{3}{2} - \frac{1}{2}\ln 2 - \frac{1}{2}$

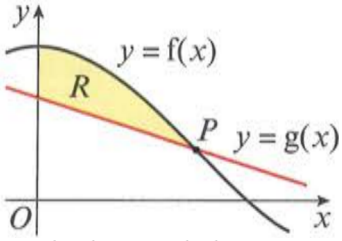
Hint

In part b, the value of the integral will be negative as the area is below the x-axis. Remember to give a positive answer for the shaded area.

Hint

To find the total area you need to consider R_1 and R_2 separately.

3. The diagram shows part of the curve $y = \mathrm{f}(x)$ and the line $y = \mathrm{g}(x)$, where $\mathrm{f}(x) = 2\cos\frac{1}{2}x + 1$ and $\mathrm{g}(x) = -\frac{1}{\pi}x + 2$.



The region R is bounded by the curve, the line and the y-axis.

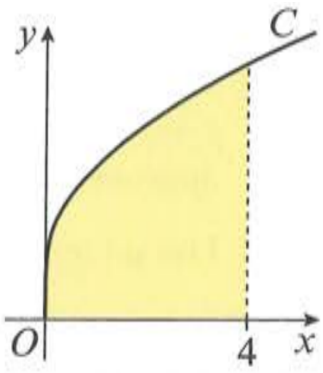
a. Show that the coordinates of point P are $(\pi, 1)$.

b. Use integration to find the exact area of R.

Hint

You could consider the total area under $y = \mathrm{f}(x)$ between $x = 0$ and $x = \pi$, then subtract the area of the trapezium under $y = \mathrm{g}(x)$. Or you could use the formula for the area between two curves: $\int_a^b (\mathrm{f}(x) - \mathrm{g}(x))\,\mathrm{d}x$

4. The curve C has parametric equations $x = t^2$, $y = 2t + 1$, $t \geq 0$.



a. Write down $\dfrac{\mathrm{d}x}{\mathrm{d}t}$

b. Find the value of t when $x = 0$.

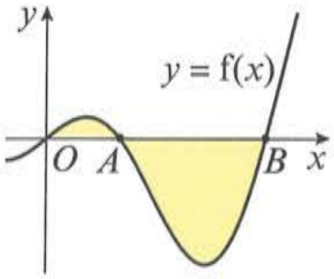
c. Explain why $t = 2$, and not -2, when $x = 4$.

d. Find the exact area of the region bounded by the curve, the x-axis, the y-axis and the line $x = 4$.

Hint

When a curve is defined parametrically, you can find the area bounded by the curve, the x-axis and the lines $x = a$ and $x = b$ using $\displaystyle\int_a^b y\frac{\mathrm{d}x}{\mathrm{d}t}\,\mathrm{d}t$

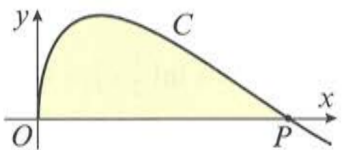
5. E/P The diagram shows part of the curve $y = \mathrm{f}(x)$ where $\mathrm{f}(x) = \dfrac{1}{2}x\cos x$.



a. Verify that points A and B have coordinates $\left(\dfrac{\pi}{2}, 0\right)$ and $\left(\dfrac{3\pi}{2}, 0\right)$ respectively. (1 mark)

b. Find the exact area of the shaded region. (7 marks)

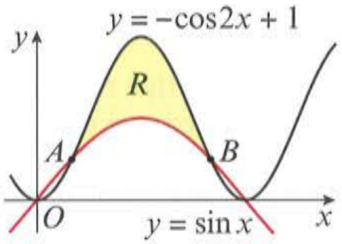
6. E/P The curve C has parametric equations $x = t^2$, $y = 4\sin t$, $t \geq 0$.



a. Find the coordinates of point P. (1 mark)

b. Find the area of the shaded region. (5 marks)

7. E/P The diagram shows part of the curves $y = \sin x$ and $y = -\cos 2x + 1$.
The region R is bounded by the two curves.

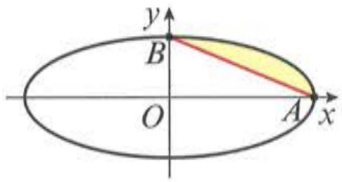


a. Using an appropriate trigonometric identity, show that point A has x-coordinate $\frac{\pi}{6}$, and find the x-coordinate of point B. (3 marks)

b. Use calculus to find the exact area of R. (6 marks)

8. E/P The diagram shows the curve with parametric equations

$$x = 12\cos\theta, \quad y = 5\sin\theta, \quad 0 \leq \theta \leq 2\pi$$

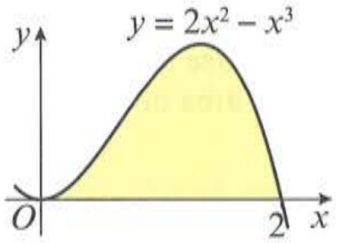


The curve intersects the positive x- and y-axes at points A and B respectively. Find the exact area of the shaded region enclosed by the curve and the line segment AB. (7 marks)

11.9 The trapezium rule

Practice Book, printed page 98.

1. The diagram shows a sketch of part of the curve with equation $y = 2x^2 - x^3$.



a. Copy and complete the table.

x	0	0.5	1	1.5	2
y	0				0

b. Explain why $h = 0.5$.

c. Use the trapezium rule, with all the y-values in the completed table, to find an approximate value for the area of the shaded region.

d. Use integration to calculate the exact value of $\int_0^2 (2x^2 - x^3)\,\mathrm{d}x$.

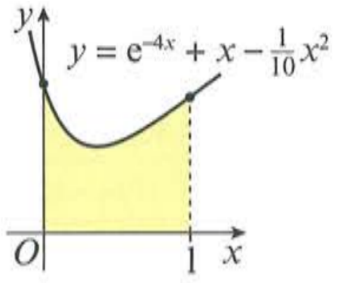
Hint

$$\int_a^b y\,\mathrm{d}x = \frac{1}{2}h(y_0 + 2(y_1 + y_2 + \ldots + y_{n-1}) + y_n)$$

where $h = \dfrac{b-a}{n}$ and $y_i = \mathrm{f}(a + ih)$.

2. The diagram shows a sketch of part of the curve with equation

$$y = e^{-4x} + x - \frac{1}{10}x^2$$



a. Copy and complete the table.

x	0	0.25	0.5	0.75	1
y	1	0.6116		0.7435	0.9183

b. Use the trapezium rule, with all the y-values in the completed table, to obtain an estimate for the area of the shaded region, giving your answer to 3 decimal places.

c. Use integration to calculate the exact value of $\int_0^1 \left(e^{-4x} + x - \frac{1}{10}x^2\right)\mathrm{d}x$.

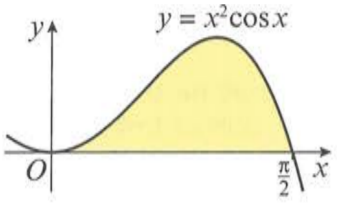
d. Use the shape of the curve in the diagram to explain why the answer in part b was an overestimate.

e. Explain how the accuracy of your approximation in part b could be improved.

Hint

The trapezium rule will overestimate a convex curve and underestimate a concave curve.

3. E The diagram shows part of the curve with equation $y = x^2 \cos x$.



a. Copy and complete the table with the corresponding y-value for $x = \frac{3\pi}{8}$

x	0	$\frac{\pi}{8}$	$\frac{\pi}{4}$	$\frac{3\pi}{8}$	$\frac{\pi}{2}$
y	0	0.142	0.436		0

(1 mark)

Given that $I = \int_0^{\frac{\pi}{2}} x^2 \cos x \, \mathrm{d}x$,

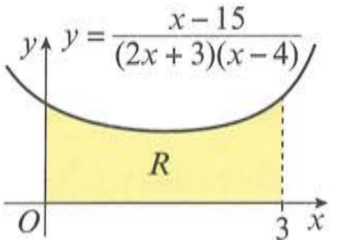
b. use the trapezium rule, with four strips to find an approximate value for I, giving your answer to 3 significant figures. (4 marks)

c. Use integration to calculate the exact value of I. (5 marks)

d. Calculate the percentage error of the approximation in part b. (1 mark)

4. E/P The diagram shows part of the curve with equation

$$y = \frac{x - 15}{(2x + 3)(x - 4)}$$



a. Copy and complete with the y-value corresponding to $x = 1.5$.

x	0	0.5	1	1.5	2	2.5	3
y	1.25	1.038	0.933		0.929	1.041	1.333

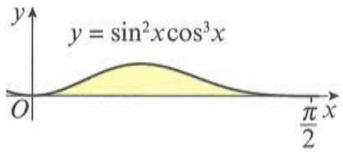
(1 mark)

b. Use the trapezium rule, with all the values from your table, to estimate the area of the region R, giving your answer to 2 decimal places. (4 marks)

c. Find the exact value of $\int_0^3 \frac{x - 15}{(2x + 3)(x - 4)} \, \mathrm{d}x$, giving your answer in the form $\ln a$, where a is an exact value to be found. (5 marks)

d. Calculate the percentage error of the approximation in part b. (1 mark)

5. E/P The diagram shows part of the curve with equation $y = \sin^2 x \cos^3 x$



a. Copy and complete the table with the corresponding y-value for $x = \frac{\pi}{8}$

x	0	$\frac{\pi}{8}$	$\frac{\pi}{4}$	$\frac{3\pi}{8}$	$\frac{\pi}{2}$
y	0		0.1768	0.0478	0

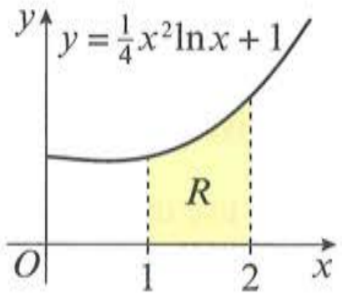
(1 mark)

Given that $I = \int_0^{\frac{\pi}{2}} \sin^2 x \cos^3 x \, \mathrm{d}x$,

b. use the trapezium rule, with four strips to find an approximate value for I, giving your answer to 4 significant figures (4 marks)

c. use a suitable trigonometric substitution to calculate the exact value of I. (5 marks)

6. E/P The diagram shows part of the curve with equation $y = \frac{1}{4}x^2 \ln x + 1$.



a. Complete the table with the y-value corresponding to $x = 1.25$.

x	1	1.25	1.5	1.75	2
y	1		1.2281	1.4285	1.6931

(1 mark)

b. Use the trapezium rule, with all the y-values from the completed table, to obtain an estimate for the area of the shaded region R. (4 marks)

c. Show by integration that the exact value of $\int_1^2 \left(\frac{1}{4}x^2 \ln x + 1\right) \mathrm{d}x$ is $\frac{2}{3}\ln 2 + \frac{29}{36}$. (5 marks)

11.10 Solving differential equations

Practice Book, printed page 100.

1. Given that $\frac{1}{(x^2+6)^5} \times \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x}{y}$

a. show that $\int y \, \mathrm{d}y = \int x(x^2+6)^5 \, \mathrm{d}x$.

b. Hence show that $\frac{1}{2}y^2 = \frac{1}{12}(x^2+6)^6 + c$.

Hint

When $\frac{\mathrm{d}y}{\mathrm{d}x} = \mathrm{f}(x)\mathrm{g}(y)$, you can write $\int \frac{1}{\mathrm{g}(y)} \mathrm{d}y = \int \mathrm{f}(x) \, \mathrm{d}x$.
This method is called separating the variables.

2. Find the general solution to the following differential equations, giving your answers in the form $y = \mathrm{f}(x)$.

a. $\dfrac{\mathrm{d}y}{\mathrm{d}x} = e^{4x-3y}$

b. $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \sin^2 x \sec y$

c. $\dfrac{\mathrm{d}y}{\mathrm{d}x} = 6xy(x+1)$

d. $\dfrac{\mathrm{d}y}{\mathrm{d}x} = e^{2y} \sin x \cos^2 x$

Hint

Use the method of separating the variables, and remember to rearrange your integrated expression so it is in the form $y = \mathrm{f}(x)$.

3. Find particular solutions to the following differential equations using the given boundary conditions.

a. $(x+1)(x+2)\dfrac{\mathrm{d}y}{\mathrm{d}x} = xy + 3y;\ x = 1,\ y = 1$

b. $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \sin^2 y + e^x \sin^2 y;\ x = \ln 2,\ y = \dfrac{\pi}{4}$

c. $\operatorname{cosec} y \cos^2 x \dfrac{\mathrm{d}y}{\mathrm{d}x} = \sin y;\ x = \dfrac{\pi}{4},\ y = \dfrac{\pi}{4}$

d. $\dfrac{2}{x} \times \dfrac{\mathrm{d}y}{\mathrm{d}x} = -\dfrac{2}{y};\ x = 12,\ y = -5$

Hint

Substitute the boundary conditions into your integrated expression to find the value of the constant c.

4.

a. Find the general solution to $\dfrac{\mathrm{d}y}{\mathrm{d}x} = 4x - 1$.

b. On the same set of axes, sketch the particular solutions to this differential equation when $c = -10$, -3 and 0, where c is the constant of integration.

Hint

The general solution provides a family of solutions. Choosing some values of c will allow you to see what this family of curves look like.

5. E/P Given that $y = 0$ when $x = 1$, find the particular solution to the differential equation $xy\dfrac{\mathrm{d}y}{\mathrm{d}x} = (1+x^2)(1-y^2)$, giving your answer in the form $y^2 = \mathrm{f}(x)$. (6 marks)

6. E/P Find the particular solution to the differential equation $\sec y \dfrac{\mathrm{d}y}{\mathrm{d}x} = 2\cot y \sin^2 x + \cot y$, with the boundary condition $y = \dfrac{\pi}{4}$ at $x = \dfrac{\pi}{4}$. Give your answer in the form $\sec y = \mathrm{f}(x)$. (8 marks)

7. E/P Find the general solution to the differential equation $e^{2y-x}\dfrac{\mathrm{d}y}{\mathrm{d}x} = xe^{2y} - 4x$, leaving your answer in the form $\ln|\mathrm{g}(y)| = \mathrm{f}(x)$. (6 marks)

8. E/P

a. Express $\dfrac{8x-11}{(2x-5)(x+2)}$ in partial fractions. (3 marks)

b. Given that $x > \dfrac{5}{2}$, find the general solution to the differential equation

$$(2x-5)(x+2)\frac{\mathrm{d}y}{\mathrm{d}x} = y(8x-11)$$

(5 marks)

c. Hence find the particular solution to the differential equation that satisfies $x = 3$, $y = 25$, giving your answer in the form $y = \mathrm{f}(x)$. (4 marks)

11.11 Modelling with differential equations

Practice Book, printed page 101.

1. The population of wild geese at time t years is given by $\frac{\mathrm{d}P}{\mathrm{d}t} = 3P^{\frac{1}{3}}$.

a. Show that $\frac{3}{2}P^{\frac{2}{3}} = 3t + c$.

Given that initially the population of wild geese is 1000,

b. find the value of the constant c.

c. Hence show that $P = (2t + 100)^{\frac{3}{2}}$.

d. Find the time taken for the population to reach 2500.

Hint

Solve the differential equation using the method of separating the variables.

2. The value, £V, of a laptop decreases at a rate proportional to the current value of the laptop.

The differential equation that models this situation is $\frac{\mathrm{d}V}{\mathrm{d}t} = -kV$, where t is the time in years since the laptop was purchased.

a. Explain:

i. the meaning of $\frac{\mathrm{d}V}{\mathrm{d}t}$ in the context of this question

ii. the significance of the negative sign

iii. the meaning of kV.

b. Solve the differential equation to show that $V = V_0 e^{-kt}$ where V_0 is the initial value of the laptop.

c. Given that initially the value of the laptop is £800, state the value of V_0.

d. Given further that after 4 years, the value of the laptop is £550, find the value of k to 2 significant figures.

Hint

You might need to form differential equations based on a proportionality relationship. Read the question carefully: in this case the **rate** of decrease is proportional to the value.

3. E/P The rate of increase of the world's population, P, in millions, t years after 1900 is proportional to the existing population.

Given that in 1900 the world's population was 1600 million and by 1950 the world's population was 2600 million,

a. show that $P = Ae^{kt}$, stating the exact values of the constants A and k. (5 marks)

b. Find the time when, according to the model, the world's population will be exactly three times the population in 1900. (2 marks)

In November 2018 the population of the world was 7700 million.

c. Evaluate the model based on this information. (1 mark)

4. E/P A bowl of hot soup, with a temperature of T°C is left to cool for t minutes in a room with a temperature of 20°C. The rate at which the soup cools is proportional to the difference in temperature between the soup and the room.

a. Explain clearly why the differential equation $\frac{\mathrm{d}T}{\mathrm{d}t} = -k(T - 20)$, where k is a positive constant, models this situation. (3 marks)

Initially the temperature of the soup is 75°C. After five minutes the soup has cooled to 65°C.

b. Find, to 1 decimal place, the time when the soup reaches a temperature of 50°C. (7 marks)

5. E/P Water enters a large cylindrical rain barrel, with diameter 60 cm, at a rate of $100\pi\,\text{cm}^3\,\text{s}^{-1}$. Water flows out of the bottom of the barrel at a rate of $3\pi h\,\text{cm}^3\,\text{s}^{-1}$.

a. Show that this situation can be modelled by the differential equation $900\dfrac{\mathrm{d}h}{\mathrm{d}t} = 100 - 3h$. (3 marks)

b. Given that initially the water level is 20 cm, find the exact time when the water level is 30 cm. (5 marks)

6. E/P Following a robbery, a circular police cordon is formed to capture the criminal. The circle has a radius of r km and an area of $A\,\text{km}^2$. The radius is gradually decreased in an effort to capture the criminal.

The rate of decrease of the area, in km^2 per minute, at time t minutes after the cordon is initially formed can be modelled as $\dfrac{\mathrm{d}A}{\mathrm{d}t} = -k\sin\left(\dfrac{t}{4\pi}\right)$, $t \geq 0$, where k is a positive constant.

a. Show that $\dfrac{\mathrm{d}r}{\mathrm{d}t} = -\dfrac{k}{2\pi r}\sin\left(\dfrac{t}{4\pi}\right)$. (2 marks)

Given that the initial radius of the cordon is 10 km and after $2\pi^2$ minutes the radius is 5 km,

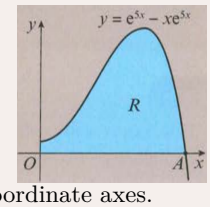
b. find an expression for r^2 in terms of t (6 marks)

c. find the time when the radius is reduced to 0 km. (3 marks)

Problem solving: Set A

Bronze

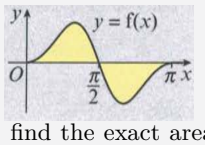
The diagram shows a sketch of the curve with equation $y = e^{5x} - xe^{5x}$, $x > 0$



The region R is bounded by the curve and the coordinate axes.
Use calculus to find the exact area of R. (5 marks)

Silver

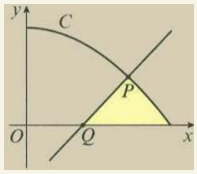
The diagram shows part of the curve with equation $y = \mathrm{f}(x)$, where $\mathrm{f}(x) = 2\sin x \cos x(1 + \sin x)^4$, $x \geq 0$



By means of a suitable substitution, or otherwise, find the exact area of the shaded region bounded by the curve and the x-axis. (6 marks)

Gold

The curve C has parametric equations
$x = 6\cos t$, $y = 4\sin^2 t$, $0 \leq t \leq \dfrac{\pi}{2}$



P is the point on the curve where $t = \dfrac{\pi}{6}$
Line PQ is the normal to C at point P.
Find the exact area of the shaded region bounded by PQ, the curve and the x-axis. (10 marks)

Problem solving: Set B

Bronze

a. Find the general solution to the differential equation $(x^2 + x + 2)\dfrac{\mathrm{d}y}{\mathrm{d}x} = 4xy + 2y$. (3 marks)

b. Find the particular solution to this differential equation that satisfies $y = 1$ at $x = 0$. (3 marks)

Silver

Find the particular solution to the differential equation $e^{2y-x}\dfrac{\mathrm{d}y}{\mathrm{d}x} = x(e^{2y} - 1)$ with boundary conditions $x = 0$, $y = \ln 2$, giving your answer in the form $\ln|\mathrm{f}(y)| = \mathrm{g}(x)$. (6 marks)

Gold

The sale of tickets for a cup final can be modelled by the differential equation
$\dfrac{\mathrm{d}T}{\mathrm{d}t} = \dfrac{1}{120000}T(80000 - T)$
where T is the number of tickets sold and t is the time in days since the tickets first went on sale.
Given that before tickets went on sale (at time $t = 0$), 16000 tickets were given away to corporate sponsors,

a. solve the differential equation giving your answer in the form $T = \dfrac{a}{b + ce^{-\frac{2}{3}t}}$ (7 marks)

b. Assuming the stadium sells out as $t \to \infty$, state the capacity of the stadium. (1 mark)

Now try this ⟶ Exam question bank Q8, Q20, Q23, Q38, Q47, Q53, Q56, Q74

Selected exam question bank questions

Question 8

Find $\int_0^{\frac{\pi}{4}} (6\sec^2 x - 4\tan^2 x)\,\mathrm{d}x$. (3 marks)

Question 20

Find $\int x^3 \ln(x^2+1)\,\mathrm{d}x$. (7 marks)

Question 23

Given that $\int_{\ln 2}^{\ln a} \frac{2e^x}{e^x+1}\,\mathrm{d}x = \ln 4$, $a > 0$, find the value of a. (4 marks)

Question 38

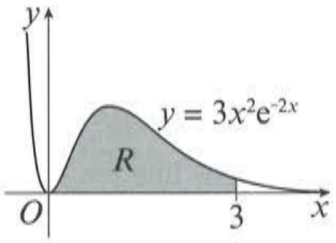
Given that k is a positive integer, show that $\int_{3k}^{4k} \frac{4}{5x+k}\,\mathrm{d}x$ is independent of k. (4 marks)

Question 47

Using the substitution $u = \operatorname{cosec} x$, or otherwise, find the exact value of $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^3 x\,\mathrm{d}x$. (8 marks)

Question 53

The diagram shows the part of the curve with equation $y = 3x^2e^{-2x}$.



a. Complete the table with the y-value corresponding to $x = 1.5$.

x	0	0.5	1	1.5	2	2.5	3
y	0	0.2759	0.4060		0.2198	0.1263	0.0669

(1 mark)

b. Use the trapezium rule, with all your y-values from the completed table, to obtain an estimate for the area of the shaded region R. (3 marks)

c. Show, by integration, that the exact value of $\int_0^3 3x^2e^{-2x}\,\mathrm{d}x$ is $\frac{3}{4} - \frac{75}{4e^6}$ (5 marks)

Question 56

Following heavy rainfall, water flows into a reservoir at a rate $200\,\mathrm{m}^3\,\mathrm{h}^{-1}$. To relieve pressure on the dam, water is released at a rate of $\frac{1}{5}V\,\mathrm{m}^3\,\mathrm{h}^{-1}$, where V is the volume of the reservoir.

a. Show that $-5\dfrac{\mathrm{d}V}{\mathrm{d}t} = V - 1000$. (2 marks)

Given that when the water is initially released, the volume of the dam is $9000\,\mathrm{m}^3$,

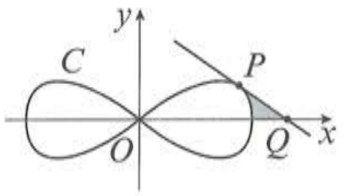
b. find the solution to the differential equation in the form $V = a + be^{-kt}$, where a, b and k are constants to be found. (7 marks)

c. State, with a reason, the minimum volume of the reservoir according to the model. (1 mark)

Question 74

The curve C has parametric equations

$$x = 6\cos t, \quad y = 2\sin 2t, \quad 0 \leq t \leq 2\pi$$



a. Find the gradient of the curve at the point P at which $t = \dfrac{\pi}{6}$ (3 marks)

b. Find the equation of the tangent to the curve at the point P. (3 marks)

c. Find the exact area of the shaded region bounded by the tangent PQ, the curve and the x-axis. (6 marks)

Answers

11.1 Integrating standard functions

1.

a. $e^x - 2\cos x + c$

b. $4\sin x + \dfrac{3}{x} + c$

c. $5\ln|x| - 2e^x + c$

d. $\dfrac{2}{3}x^{\frac{3}{2}} - \dfrac{3}{4}\cos x + c$

e. $\dfrac{11}{3}x^3 + \dfrac{1}{2}e^x - 6\ln|x| + c$

f. $3\sin x - 2\sqrt{x} + c$

2.

a. $-\cot x + c$

b. $\sec x + c$

c. $-\operatorname{cosec} x + c$

d. $4\tan x + c$

e. $-4\cot x + c$

f. $8\operatorname{cosec} x + c$

3.

a. $\tan x + \sec x + c$

b. $\dfrac{7}{2}x^2 - 5\ln x + c$

c. $3e^x + 5\cot x + c$

d. $\dfrac{1}{2}x^2 - x + \ln x + c$

4.

a. 15

b. $\ln\dfrac{5}{2} - 117$

c. $4e^{\frac{\pi}{3}} - 3$

d. $5 - \sqrt{2}$

e. 20

5. $5 + \ln\dfrac{9}{4}$

6. $p = 4$

7. $3e^{\frac{\pi}{6}} - 3e^{\frac{\pi}{3}} = 3(e^{\frac{\pi}{6}} - e^{\frac{\pi}{3}})$

8. $[2x + 3\ln x]_{2a}^{5a} = 6a + 3\ln\dfrac{5}{2}$

$\Rightarrow 6a + 3\ln\dfrac{5}{2} = \ln 1000 = 3\ln 10$

$\Rightarrow 6a = 3\ln 4$

$\Rightarrow a = \dfrac{1}{2}\ln 4 = \ln 2$

11.2 Integrating f$(ax + b)$

1.

a.

i. $16(4x - 5)^3$

ii. $-12(3 - 2x)^5$

b.

i. $(4x - 5)^4 + c$

ii. $\frac{1}{16}(4x - 5)^4 + c$

iii. $(3 - 2x)^6 + c$

iv. $-\frac{1}{6}(3 - 2x)^6 + c$

2.

a. $2e^{3x} + c$

b. $\frac{1}{3}\sin(3x + 2) + c$

c. $\frac{1}{24}(4x + 1)^6 + c$

d. $-\frac{1}{4}\operatorname{cosec} 4x + c$

e. $\frac{1}{8}\tan(8x - 1) + c$

f. $\frac{1}{5}\cot(1 - 5x) + c$

g. $\frac{5}{2}\ln|2x - 3| + c$

h. $-11e^{4-x} + c$

3.

a. $-\frac{1}{4(4x - 3)} - \frac{1}{12}(4x - 3)^3 + c$

b. $\frac{1}{10}e^{5x} - \frac{2}{3}\tan 3x + c$

c. $\frac{1}{4}e^{4x} - 2x - \frac{1}{4}e^{-4x} + c$

d. $-\frac{5}{2}\cot 2x + \frac{3}{2}\operatorname{cosec} 2x + c$

4.

a. $\ln\frac{7}{3}$

b. $\sqrt{2} - 1$

c. $16\ln 4 - \frac{33}{2}$

d. $1 - \arctan\left(\frac{\pi}{8}\right)$

5. 4

6. $2\frac{\ln 25}{\ln 7}$

7. $54 + \ln 256$

8. $k = -\frac{1}{9}$

11.3 Using trigonometric identities

1.

a. $\frac{1}{2}x - \frac{1}{4}\sin 2x + c$

b. $-3\cot x - 3x + c$

c. $\frac{3}{2}x + 2\sin x + \frac{1}{4}\sin 2x + c$

d. $\tan x - x + c$

2.

a. $\sqrt{3} - 1 - \frac{\pi}{6}$

b. $\frac{10 - 3\pi}{4}$

c. $\frac{4\sqrt{3} - \pi}{3}$

d. $4 - 2\sqrt{2} - \frac{\pi}{4}$

3.

a.

i. $\cos 5x \cos 2x - \sin 5x \sin 2x$

ii. $\cos 5x \cos 2x + \sin 5x \sin 2x$

b. L.H.S. $= \cos 7x + \cos 3x$
R.H.S. $= 2\cos 5x \cos 2x$

c. $\frac{2}{7}\sin 7x + \frac{2}{3}\sin 3x + c$

4. $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \cos^2 x \, \mathrm{d}x = \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \left(\frac{1}{2}\cos 2x + \frac{1}{2}\right) \mathrm{d}x$

$= \left[\frac{1}{4}\sin 2x + \frac{1}{2}x\right]_{\frac{\pi}{6}}^{\frac{\pi}{2}}$

$= \left(0 + \frac{\pi}{4}\right) - \left(\frac{\sqrt{3}}{2} + \frac{\pi}{12}\right)$

$= \frac{\pi}{6} - \frac{\sqrt{3}}{8}$

5. Use the trigonometric identity $\tan^2 2x + 1 = \sec^2 2x$ to obtain $\int_{\frac{\pi}{12}}^{\frac{\pi}{6}} (\sec^2 2x - 1) \, \mathrm{d}x = \left[\frac{1}{2}\tan 2x - x\right]_{\frac{\pi}{12}}^{\frac{\pi}{6}}$

$= \left(\frac{\sqrt{3}}{2} - \frac{\pi}{6}\right) - \left(\frac{1}{2\sqrt{3}} - \frac{\pi}{12}\right)$

$= \frac{\sqrt{3}}{3} - \frac{\pi}{12}$

6.

a. $\sin 6x \cos x + \cos 6x \sin x = \sin 7x \quad (1)$
$\sin 6x \cos x - \cos 6x \sin x = \sin 5x \quad (2)$
Adding (1) and (2): $2\sin 6x \cos x = \sin 7x + \sin 5x$

b. $-\frac{3}{14}\cos 7x - \frac{3}{10}\cos 5x + c$

11.4 Reverse chain rule

1.

a.

i. $6x(x^2-4)^2$

ii. $8(x+2)(x^2+4x-5)^3$

b.

i. $(x^2-4)^3+c$

ii. $\frac{5}{2}(x^2-4)^3+c$

iii. $-\frac{4}{3}(x^2+4x-5)^3+c$

2.

a. $2(x^2-7)^{\frac{3}{2}}+c$

b. $-\frac{5}{2}e^{\cos 2x}+c$

c. $\frac{1}{5}(4x^3+16x-5)^{\frac{5}{2}}+c$

d. $\frac{2}{3}(1+\tan x)^{\frac{3}{2}}+c$

e. $-\frac{1}{3}e^{\cot 3x}+c$

f. $-\frac{1}{25}\cos^5 5x+c$

3.

a. $-2\ln|2-x^2|+c$

b. $\frac{1}{3}\ln|4-3\cos x|+c$

c. $-\frac{1}{2}\ln|14+9x-6x^3|+c$

d. $-2\ln|\cos x|+c$

e. $\frac{1}{8}\ln|4-\cos 2x|+c$

f. $-\frac{1}{3}\ln|1+e^{-3x}|+c$

4.

a. $\frac{1}{12}$

b. $\frac{1}{3}\ln 2$

c. $2\ln\frac{28}{9}$

d. $\frac{1}{3}(e-e^{\sqrt{2}})$

5. $3+4\sqrt{3}$

6.

a. $\sin^5 x \equiv \sin x(\sin^2 x)(\sin^2 x)$
$\equiv \sin x(1-\cos^2 x)(1-\cos^2 x)$
$\equiv \sin x(1-2\cos^2 x+\cos^4 x)$
$\equiv \sin x-2\sin x\cos^2 x+\sin x\cos^4 x$

b. $-\frac{1}{5}\cos^5 x+\frac{2}{3}\cos^3 x-\cos x+c$

7. $32\sqrt{3}$

8. $\frac{1}{4}(e-1)$

9. $\frac{19}{648}$

10. $k = \sqrt{6}$

11.5 Integration by substitution

1.

a. 1

b. $2(u-2)$

c. Substitute $2x = 2(u-2)$, $(x+2)^4 = u^4$ and $\mathrm{d}x = \mathrm{d}u$

d. $\displaystyle\int 2(u-2)u^4\,\mathrm{d}u = \int (2u^5 - 4u^4)\,\mathrm{d}u = \frac{1}{3}u^6 - \frac{4}{5}u^5 + c$

e. $\frac{1}{3}(x+2)^6 - \frac{4}{5}(x+2)^5 + c$

2.

a. $\frac{1}{6}(x-8)^6 + \frac{8}{5}(x-8)^5 + c$

b. $-\frac{1}{7}(1+\cos x)^7 + \frac{1}{6}(1+\cos x)^6 + c$

c. $-\frac{2}{5}(2+\cot x)^{\frac{5}{2}} + \frac{4}{3}(2+\cot x)^{\frac{3}{2}} + c$

d. $\sqrt{x^2+1} + c$

3.

a. $\frac{16}{15}$

b. $8\ln\frac{4}{5}$

c. -1

d. $2 - \ln 4$

4. $\frac{2}{15} + \frac{10\sqrt{5}}{3}$

5. $\ln\frac{7}{3}$

6.

a. $x = \tan\theta$, $\frac{\mathrm{d}x}{\mathrm{d}\theta} = \sec^2\theta$

$$\begin{aligned}\int \frac{1}{1+x^2}\,\mathrm{d}x &= \int \frac{1}{1+\tan^2\theta}\sec^2\theta\,\mathrm{d}\theta \\ &= \int \frac{1}{\sec^2\theta}\sec^2\theta\,\mathrm{d}\theta \\ &= \int 1\,\mathrm{d}\theta \\ &= \theta + c \\ &= \arctan x + c\end{aligned}$$

b. $\frac{\pi}{12}$

7. $\displaystyle\int_0^4 \frac{5x}{\sqrt{2x+1}}\,\mathrm{d}x = \int_1^3 \frac{5}{u}\left(\frac{u^2-1}{2}\right)u\,\mathrm{d}u = \frac{5}{2}\left[\frac{u^3}{3} - u\right]_1^3 = \frac{50}{3}$

8. $4\sin x - 4\ln|1+\sin x| + c$

11.6 Integration by parts

1.

a. $x\sin x + \cos x + c$

b. $\ln|\cos x| + x\tan x + c$

c. $\frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x} + c$

d. $\frac{1}{16}\ln|\sin 4x| - \frac{1}{4}x\cot 4x + c$

e. $-\frac{1}{4}xe^{-4x} - \frac{1}{16}e^{-4x} + c$

f. $-\frac{1}{5}x\cos 5x + \frac{1}{25}\sin 5x + c$

2.

a. $\frac{1}{2}x^2\ln x - \frac{1}{4}x^2 + c$

b. $\frac{1}{4}x^4\ln x - \frac{1}{16}x^4 + c$

c. $x\ln x - x + c$

3.

a. $-\frac{1}{2}x^2\cos 2x + \frac{1}{2}x\sin 2x + \frac{1}{4}\cos 2x + c$

b. $-\frac{1}{27}(9x^2+6x+2)e^{-3x} + c$

c. $\frac{1}{2}x^2(1+x)^4 - \frac{1}{5}x(1+x)^5 + \frac{1}{30}(1+x)^6 + c$

4.

a. $\frac{\sqrt{2}}{8}(4-\pi)$

b. $\frac{78}{5}$

c. π

d. $\frac{\pi\sqrt{3}}{12} - \frac{1}{4}\ln 2$

e. $-\frac{\pi}{32}$

f. $6 - \frac{78}{e^3}$

5. $12\ln 8 - \frac{3}{4}\ln 2 - \frac{45}{8}$

6. $\left[\frac{1}{4}(2x^2-2x+1)e^{2x}\right]_1^2$
$= \frac{1}{4}(5)e^4 - \frac{1}{4}(1)e^2$
$= \frac{e^2}{4}(5e^2-1)$

7.

a. $-\frac{1}{\pi}x\cos\pi x + \frac{1}{\pi^2}\sin\pi x + c$

b. $\frac{17\pi^2-8}{2\pi^3}$

8. $\frac{\pi}{2} - \frac{\pi^2}{8} + \ln 2$

9.

a. $\frac{1}{2}x\sin 2x + \frac{1}{4}\cos 2x + c$

b. $\frac{1}{4}x^2 - \frac{1}{4}x\sin 2x - \frac{1}{8}\cos 2x + c$

11.7 Partial fractions

1.

a. $A = \frac{10}{3}$, $B = \frac{14}{3}$

b. $\frac{10}{3}\ln|x+2| + \frac{14}{3}\ln|x-4| + c$

2.

a. $\ln|x+5| + 3\ln|x+2| + c$

b. $-4\ln|x-2| - 3\ln|x+3| + c$

c. $6\ln|x-1| - 5\ln|x+2| + c$

d. $\frac{6}{7}\ln|x-4| - \frac{6}{7}\ln|2x-1| + c$

3.

a. $(x+4)(2x-1) = 2x^2 + 7x - 4$

$$\begin{array}{r|l} 2x^2+7x-4 & \begin{array}{r} -1 \\ -2x^2 \qquad +14 \\ \hline -2x^2-7x+4 \\ \hline 7x+10 \end{array} \end{array}$$

So $\mathrm{f}(x) = \frac{7x+10}{(x+4)(2x-1)} - 1$

b. $A = 2$, $B = 3$

c. $2\ln|x+4| + \frac{3}{2}\ln|2x-1| - x + c$

4.

a. Equate coefficients in
$16x^2 - 37x + 17 = A(2x-3)^2 + B(4-x)(2x-3) + C(4-x)$
to get simultaneous equations:
$4A - 2B = 16$
$-12A + 11B - C = -37$
$9A - 12B + 4C = 17$
Then solve to find $A = 5$, $B = 2$ and $C = -1$.

b. $-5\ln|4-x| + \ln|2x-3| + \frac{1}{2(2x-3)} + c$

5.

a. $\frac{6}{x-1} - \frac{3}{x+2}$

b. $\ln\frac{4096}{125}$

6.

a. $A = 1$, $B = -2$, $C = 1$

b. $2 + \ln\frac{5}{3}$

7.

a. $\dfrac{12}{5(2x+1)} - \dfrac{6}{5(x+3)}$

b. $\dfrac{6}{5}\ln\dfrac{4}{3}$

8.

a. $A = 1$, $B = -\dfrac{5}{2}$, $C = \dfrac{5}{2}$

b. $2 + \dfrac{5}{2}\ln\dfrac{9}{5}$

11.8 Finding areas

1. $\dfrac{8}{5}$

2.

a. $x\ln\dfrac{1}{2}x = 0 \Rightarrow x = 0$ or $\ln\dfrac{1}{2}x = 0$

So A is where $\ln\dfrac{1}{2}x = 0$

$\Rightarrow \dfrac{1}{2}x = 1 \Rightarrow x = 2$

So A has coordinates $(2, 0)$.

b. $R_1 = -\displaystyle\int_1^2 x\ln\left(\frac{1}{2}x\right)\mathrm{d}x = -\left[\frac{1}{2}x^2\ln\left(\frac{1}{2}x\right) - \frac{1}{4}x^2\right]_1^2 = \frac{3}{4} - \frac{1}{2}\ln 2$

c. $R_2 = \displaystyle\int_2^3 x\ln\left(\frac{1}{2}x\right)\mathrm{d}x = \left[\frac{1}{2}x^2\ln\left(\frac{1}{2}x\right) - \frac{1}{4}x^2\right]_2^3 = \frac{9}{2}\ln\frac{3}{2} - \frac{5}{4}$

Total area $= R_1 + R_2 = \dfrac{9}{2}\ln\dfrac{3}{2} - \dfrac{1}{2}\ln 2 - \dfrac{1}{2}$

3.

a. $\mathrm{f}(\pi) = 2\cos\dfrac{1}{2}\pi + 1 = 1$

$\mathrm{g}(\pi) = -\dfrac{\pi}{\pi} + 2 = 1$

b. $4 - \dfrac{\pi}{2}$

4.

a. $2t$

b. $t = 0$

c. $t \neq -2$ because y is positive and $2(-2) + 1 = -3$

d. $\dfrac{44}{3}$

5.

a. $\dfrac{1}{2}\left(\dfrac{\pi}{2}\right)\cos\left(\dfrac{\pi}{2}\right) = 0$, $\dfrac{1}{2}\left(\dfrac{3\pi}{2}\right)\cos\left(\dfrac{3\pi}{2}\right) = 0$

b. $\dfrac{5\pi}{4} - \dfrac{1}{2}$

6.

a. $(\pi^2, 0)$

b. 8π

7.

a. $\sin x = -\cos 2x + 1$

$\Rightarrow \sin x = -(1 - 2\sin^2 x) + 1$

$\Rightarrow 2\sin^2 x - \sin x = 0$

$\Rightarrow \sin x(2\sin x - 1) = 0$

$2\sin x = 1 \Rightarrow x = \frac{\pi}{6}$ or $\frac{5\pi}{6}$

So the x-coordinate of B is $\frac{5\pi}{6}$

b. $\frac{2\pi}{3} - \frac{\sqrt{3}}{2}$

8. $15\pi - 30$

11.9 The trapezium rule

1.

a.

x	0	0.5	1	1.5	2
y	0	0.375	1	1.125	0

b. There are 4 trapezia and each has a height of 0.5 as $4 \times 0.5 = 2 = b - a$

c. 1.25

d. $\frac{4}{3}$

2.

a.

x	0	0.25	0.5	0.75	1
y	1	0.6116	0.6103	0.7435	0.9183

b. 0.731

c. $\frac{43}{60} - \frac{1}{4e^4}$

d. The graph is convex, so the lines connecting the two endpoints would be above the curve, giving a larger answer than the real answer.

e. Decrease the width of each strip by increasing the number of values used.

3.

a.

x	0	$\frac{\pi}{8}$	$\frac{\pi}{4}$	$\frac{3\pi}{8}$	$\frac{\pi}{2}$
y	0	0.142	0.436	0.531	0

b. 0.436

c. $\frac{\pi^2}{4} - 2$

d. 6.8%

4.

a.

x	0	0.5	1	1.5	2	2.5	3
y	1.25	1.038	0.933	0.9	0.929	1.041	1.333

b. 3.07

c. $\ln(12\sqrt{3})$

d. 1.2%

5.

a.

x	0	$\frac{\pi}{8}$	$\frac{\pi}{4}$	$\frac{3\pi}{8}$	$\frac{\pi}{2}$
y	0	0.1155	0.1768	0.0478	0

b. 0.1336

c. $\dfrac{2}{15}$

6.

a.

x	1	1.25	1.5	1.75	2
y	1	1.0872	1.2281	1.4285	1.6931

b. 1.273 (3 d.p.)

c. $\displaystyle\int \left(\frac{1}{4}x^2 \ln x + 1\right) \mathrm{d}x = \frac{1}{12}x^3 \ln x - \frac{1}{36}x^3 + x + c$

Substitute in the limits and simplify to obtain $\dfrac{2}{3}\ln 2 + \dfrac{29}{36}$

11.10 Solving differential equations

1.

a. $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{x}{y} \times (x^2+6)^5 \Rightarrow \displaystyle\int y \,\mathrm{d}y = \int x(x^2+6)^5 \,\mathrm{d}x$

b. Using the substitution $u = x^2 + 6$,

$\displaystyle\int y \,\mathrm{d}y = \frac{1}{2}\int u^5 \,\mathrm{d}u$

$\Rightarrow \dfrac{1}{2}y^2 = \dfrac{1}{12}u^6 + c = \dfrac{1}{12}(x^2+6)^6 + c$

2.

a. $y = \dfrac{4x}{3} + c$

b. $y = \arcsin\left(\dfrac{1}{2}x - \dfrac{1}{4}\sin 2x + c\right)$

c. $y = Ae^{2x^3+3x^2}$

d. $y = -\dfrac{1}{2}\ln\left|\dfrac{2}{3}\cos^3 x + c\right|$

3.

a. $y = \dfrac{3(x+1)^2}{4(x+2)}$

b. $y = \operatorname{arccot}(3 + \ln 2 - e^x - x)$

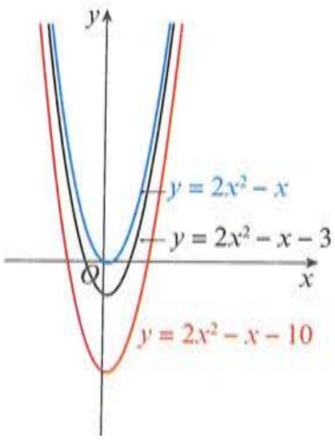
c. $\tan x + \cot y - 2 = 0$

d. $x^2 + y^2 = 13^2$

4.

a. $y = 2x^2 - x + c$

b.



5. $y^2 = 1 - \frac{1}{x^2}e^{-x^2+1}$

6. $\sec y = 2x - \frac{1}{2}\sin 2x + \sqrt{2} + \frac{1}{2} - \frac{\pi}{2}$

7. $\ln|e^{2y} - 4| = 2e^x(x-1) + c$

8.

a. $\frac{2}{2x-5} + \frac{3}{x+2}$

b. $y = A(2x-5)(x+2)^3$

c. $y = \frac{1}{5}(2x-5)(x+2)^3$

11.11 Modelling with differential equations

1.

a. Separate the variables: $\int P^{-\frac{1}{3}}\,\mathrm{d}P = \int 3\,\mathrm{d}t$ and integrate to obtain $\frac{3}{2}P^{\frac{2}{3}} = 3t + c$.

b. $c = 150$

c. $\frac{3}{2}P^{\frac{2}{3}} = 3t + 150 \Rightarrow P^{\frac{2}{3}} = 2t + 100 \Rightarrow P = (2t+100)^{\frac{3}{2}}$

d. 42.1 years

2.

a.

i. The rate of change of the value of the laptop with respect to time.

ii. The value of the laptop is decreasing

iii. Proportional to the existing value of the laptop.

b. $\int \frac{1}{V}\,\mathrm{d}V = \int -k\,\mathrm{d}t \Rightarrow \ln V = -kt + c \Rightarrow V = e^{-kt+c}$
$\Rightarrow V = Ae^{-kt}$
When $t = 0$, $V = V_0 = Ae^{-k\times 0}$, so $A = V_0$ and $V = V_0e^{-kt}$

c. 800

d. $k = 0.094$

3.

a. $\frac{\mathrm{d}P}{\mathrm{d}t} = kP \Rightarrow \int \frac{1}{P}\,\mathrm{d}P = \int k\,\mathrm{d}t$
$\Rightarrow \ln P = kt + c \Rightarrow P = e^{kt+c}$
$\Rightarrow P = Ae^{kt}$

$A = 1600,\ k = \dfrac{1}{50}\ln\dfrac{13}{8}$

b. 2013

c. The model is clearly inaccurate for larger values of t.

4.

a. $\dfrac{\mathrm{d}T}{\mathrm{d}t}$ is the rate of change of the temperature of the soup with respect to time.
$(T - 20)$ is the difference between the temperature of the soup and the room temperature.
$-k$ signifies the soup cooling.

b. 15.1 minutes

5.

a. $\dfrac{\mathrm{d}h}{\mathrm{d}t} = \dfrac{100\pi - 3\pi h}{900\pi}$

$900\dfrac{\mathrm{d}h}{\mathrm{d}t} = 100 - 3h$

b. $300\ln 4$ seconds

6.

a. $\dfrac{\mathrm{d}A}{\mathrm{d}r} = 2\pi r$

$\dfrac{\mathrm{d}r}{\mathrm{d}t} = \dfrac{\mathrm{d}r}{\mathrm{d}A} \times \dfrac{\mathrm{d}A}{\mathrm{d}t}$

$= \dfrac{1}{2\pi r} \times \left(-k\sin\left(\dfrac{t}{4\pi}\right)\right)$

$= -\dfrac{k}{2\pi r}\sin\left(\dfrac{t}{4\pi}\right)$

b. $r^2 = 75\cos\left(\dfrac{t}{4\pi}\right) + 25$

c. 24.0 minutes

Problem solving: Set A

Bronze $\dfrac{1}{25}(e^5 - 6)$

Silver $\dfrac{86}{5}$

Gold $16 - \dfrac{26}{3}\sqrt{3}$

Problem solving: Set B

Bronze

a. $y = k(x^2 + x + 2)^2$

b. $y = \dfrac{(x^2 + x + 2)^2}{4}$

Silver $\ln|e^{2y} - 1| = 2e^x(x - 1) + 2 + \ln 3$

Gold

a. $T = \dfrac{80000}{1 + 4e^{-\frac{2}{3}t}}$

b. 80000

Selected exam question bank answers

Question 8 $\pi + 2$

Question 20 $\frac{1}{4}x^4 \ln(x^2+1) - \frac{1}{4}\ln(x^2+1) - \frac{1}{8}x^2(x^2-2) + c$

Question 23 $a = 5$ or -7

Question 38 Integral $= \frac{4}{5}\ln\frac{21}{16}$. Answer is independent of k, as k does not appear in the solution.

Question 47 $\frac{1}{2}(1 - \ln 2)$

Question 53

a.

x	0	0.5	1	1.5	2	2.5	3
y	0	0.2759	0.4060	0.3361	0.2198	0.1263	0.0669

b. 0.699 (3 s.f.)

c. $\int_0^3 3x^2 e^{-2x}\,\mathrm{d}x = \left[-\frac{3}{4}(2x^2+2x+1)e^{-2x}\right]_0^3$
$= \frac{3}{4} - \frac{75}{4e^6}$

Question 56

a. $\frac{\mathrm{d}V}{\mathrm{d}t} = 200 - \frac{1}{5}V$
Multiply by -5 to obtain
$-5\frac{\mathrm{d}V}{\mathrm{d}t} = V - 1000$

b. $V = 1000 + 8000e^{-\frac{1}{5}t}$

c. $1000\,\mathrm{m}^3$. As $t \to \infty$, $8000e^{-\frac{1}{5}t} \to 0$, therefore $V \to 1000$.

Question 74

a. $-\frac{2}{3}$

b. $2x + 3y = 9\sqrt{3}$

c. $\frac{5}{4}$